

ENVIRONMENT ACTION PLAN MAKINDYE SSABAGABO MUNICIPALITY 2025-2030

Approved by the Natural Resource Committee under minute number ENRC/004/06/2025

GREATER KAMPALA METROPOLITAN AREA URBAN DEVELOPMENT PROGRAM

THE WORLD BANK

Executive Summary

The Makindye Ssabagabo Environmental Action Plan (EAP) provides a five-year roadmap (2026–2030) for addressing the municipality's pressing environmental challenges and building a sustainable, climate-resilient future. Developed in alignment with Uganda's National Environment Act, the National Development Plan, and the Makindye Ssabagabo Municipal Development Strategy, the EAP integrates the State of the Environment (SoE) findings into actionable priorities across six result areas: waste management, water resource protection, urban greening and biodiversity, pollution control, climate resilience, and community awareness and capacity building.

The plan responds to critical issues identified in the SoE report, including rapid urbanization, poor waste management systems, wetland degradation, biodiversity loss, water pollution, unregulated emissions, weak enforcement of environmental bylaws, and limited institutional capacity. High population growth and increased economic activity have intensified land-use pressures, while climate change impacts—such as flooding, drought, and rising temperatures—threaten livelihoods and municipal infrastructure.

The EAP sets clear outcomes for each result area, supported by specific outputs and actions. These include upgrading waste collection fleets, establishing material recovery facilities, mapping and protecting wetland boundaries, restoring green spaces, installing air quality monitoring stations, developing climate change adaptation strategies, and creating community environmental watch committees. Emphasis is placed on integrating circular economy principles, promoting sustainable livelihoods, and enforcing environmental governance through bylaws, monitoring, and partnerships with civil society, the private sector, and national agencies such as NEMA and MWE.

Implementation will be guided by a results-based framework that links actions to measurable indicators, timelines, and responsible institutions. Short-term priorities (2026–2027) focus on enforcement, awareness-raising, and foundational infrastructure such as composting sites and monitoring systems. Medium- to long-term interventions (2028–2030) emphasize ecosystem restoration, climate adaptation, and institutional capacity strengthening.

The plan's estimated budget of UGX 12 billion over five years is distributed across the six result areas, with funding sourced from municipal allocations, national government transfers, development partners, and private sector investment. Resource mobilization strategies prioritize leveraging external funding, establishing an Environment Fund, and integrating environmental levies to support long-term sustainability.

The EAP for Makindye Ssabagabo Municipality aims to improve environmental quality, safeguard natural resources, reduce vulnerability to climate change, and enhance the health, well-being, and economic opportunities of its residents. The plan recognizes that environmental management is a shared responsibility and calls for strong community engagement, cross-sectoral collaboration, and sustained political commitment to achieve its vision of a clean, green, and resilient municipality.

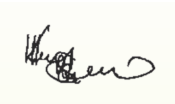
Foreword

The Makindye Ssabagabo Environment Action Plan 2025–2030 marks a pivotal step in our municipality's journey towards a cleaner, greener, and more climate-resilient future. In recent years, Makindye Ssabagabo has experienced rapid urban growth and economic expansion, bringing with it both opportunities and unprecedented environmental pressures. Wetland degradation, waste management challenges, pollution, biodiversity loss, and the growing impacts of climate change threaten the natural ecosystem that sustains our lives and livelihoods. This plan responds to these challenges by offering a comprehensive, evidence-based, and results-oriented framework for action.

This Environment Action Plan outlines clear priorities across six strategic result areas: waste management, water resource protection, urban greening and biodiversity conservation, pollution control, climate resilience, and community awareness and capacity building. It aligns with the national legislation, global environmental commitments, and our municipal development strategy.

I extend my sincere appreciation to all stakeholders whose commitment, time, and expertise have shaped this plan. Special thanks go to the World Bank, whose support through the Greater Kampala Metropolitan Urban Development Program has been instrumental in enabling the research, stakeholder engagement, and technical work that underpin this document.

In conclusion, I urge all residents, institutions, and partners to take full ownership of this Environmental Action Plan and actively contribute to its implementation. By working together, we can transform these commitments into tangible results that benefit our municipality today and for generations to come.



Ssemwanga Godfrey

His Worship, Mayor

Makindye Ssabagabo Municipality

Acknowledgement

The development of the Makindye Ssabagabo Environment Action Plan (2025–2030) would not have been possible without the invaluable support, collaboration, and commitment of numerous individuals and institutions.

First and foremost, we extend our sincere appreciation to the political leadership of Makindye Ssabagabo Municipal Council for their unwavering commitment and for creating an enabling environment for this process to take place. We particularly thank the Office of the Mayor, the Speaker, and all Municipal Councilors for their strategic guidance and support.

Special thanks go to the Department of Natural Resources for leading the technical coordination of this process and for ensuring that all stakeholder voices were meaningfully included.

We also recognize the valuable contributions from community leaders, private sector actors, and residents who participated in the consultations across the municipality. Your shared experiences and insights shaped the priorities and strategies outlined in this plan.

I wish to extend our heartfelt appreciation to the World Bank for their generous support through the Greater Kampala Metropolitan Area Urban Development Program (GKMA-UDP). The funding provided under this program was instrumental in facilitating the development and finalization of this Environment Action Plan.

Finally, we appreciate the contributions of Ardent Services International Limited, which supported the documentation and compilation of this Environment Action Plan.



Richard K Monday

Town Clerk

Makindye Ssabagabo Municipality

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Acronyms

CBD	Convention on Biological Diversity
EAP	Environment Action Plan
MEO	Municipal Environmental Officer
MWE	Ministry of Water and Environment
NCDC	Nationally Determined Contribution
NDP	National Development Plan
NEMA	National Environment Management Authority
SDGS	Sustainable Development Goals
SoE	State of the Environment Report
UNCCD	UN Convention to Combat Desertification
UNDP	United Nations Development Program
UNFCCC	United Nations Framework Convention on Climate Change

1. INTRODUCTION

1.1 Background to the Environmental Action Plan

Makindye Ssabagabo Municipality, located in Wakiso District and bordering Lake Victoria, is one of Uganda's rapidly urbanizing municipalities. Its location along key transport corridors and proximity to Kampala has fueled rapid population growth, commercial expansion, and residential development. While these trends have brought economic opportunities, they have also intensified environmental pressures, including wetland encroachment, pollution of water resources, uncontrolled solid waste accumulation, loss of biodiversity, and climate-related risks such as flooding.

The municipality is home to several ecologically significant features, including riverbanks, wetlands such as Lufuka, Kalidubi, and Kyetinda and part of the Lake Victoria shoreline. These ecosystems play a critical role in regulating water flow, providing habitats for biodiversity, supporting livelihoods, and contributing to climate resilience. However, urban expansion, inadequate waste management systems, unregulated sand mining, and weak enforcement of environmental laws have degraded many of these resources.

The Environmental Action Plan (EAP) is therefore developed as a strategic and operational framework to guide coordinated action towards the sustainable management of Makindye Ssabagabo's environment. It integrates locally identified environmental priorities with national policy directives and global sustainability commitments, ensuring that development in the municipality remains environmentally sound, socially inclusive, and economically viable.

1.2 Linkage to the State of the Environment Report

This Environmental Action Plan is directly informed by the State of the Environment Report (SoE) for Makindye Ssabagabo, which provides a comprehensive baseline assessment of the municipality's environmental status across key thematic areas including water resources, wetlands, biodiversity, waste management, climate change, disaster risk, land use and soils, urbanization and settlement patterns, and the socio-economic state.

The SoE highlights critical trends and challenges such as:

Continued wetland and riverbank degradation due to settlement expansion and agriculture.

High levels of organic and industrial pollution in surface water bodies,

Poor solid waste collection coverage, especially in informal settlements,

Increasing vulnerability to climate-related disasters, particularly floods and,

Loss of tree cover and biodiversity due to land conversion and habitat destruction.

The findings and recommendations of the SoE have been translated into the actionable results, outputs, and activities in this EAP. This is to ensure that interventions are evidence-based, targeted at priority problem areas, and aligned with both local realities and broader environmental management frameworks.

1.3 Purpose and Scope of the EAP

The purpose of the Makindye Ssabagabo Environmental Action Plan is to provide a structured, results-based approach for addressing the municipality's environmental challenges over the period 2026–2030. It serves as a roadmap for planning, implementing, monitoring, and evaluating interventions that promote sustainable environmental management while supporting socio-economic development.

The scope of the EAP covers six priority result areas identified through consultations and technical assessments:

1. Waste Management – improving collection, recycling, and disposal systems.
2. Water Resource Protection – safeguarding wetlands, rivers, and other water sources from pollution and encroachment.
3. Urban Greening and Biodiversity Protection – enhancing green spaces and conserving habitats.
4. Pollution Control – regulating air, land, and water pollutants.
5. Climate Resilience – strengthening the municipality's ability to anticipate, prepare for, and respond to climate-related risks.
6. Community Awareness and Capacity Building – empowering local communities, schools, and institutions to take an active role in environmental stewardship.

This EAP will guide all environmental actions undertaken by the municipality, divisions, community-based organizations, and partner agencies. It outlines clear responsibilities, timeframes, and resources required, ensuring coordination and accountability in delivery. The plan is also a tool for mobilizing financial and technical support from government agencies, the private sector, development partners, and civil society.

1.4 Legal and Policy Frameworks

Makindye Ssabagabo Municipality operates within a complex, multi-tiered legal and policy framework that shapes its environmental management strategies. These frameworks span international conventions, national legislation, and local policies—each offering distinct mandates, principles, and operational guidance. Together, they form the backbone of the municipality's response to environmental challenges such as wetland degradation, climate hazards, and urban expansion.

The table below (Table 1) presents the key legal and policy instruments relevant to environmental management in Makindye Ssabagabo. It highlights their scope and explains how each framework informs the municipality's planning, implementation, and monitoring efforts. These frameworks translate into both obligations and strategic opportunities. The National Environment Act (2019) mandates the preparation of biennial State of Environment Reports, positioning the municipality as a key actor in Uganda's environmental accountability system. Meanwhile, the National Wetlands Policy (1995) provides essential guidance for safeguarding critical ecosystems such as Kyetinda, and Lufuka wetlands, which face increasing pressure from urban encroachment.

At a broader level, Uganda Vision 2040 and the Fourth National Development Plan (NDP IV) offer a strategic blueprint for harmonizing urban growth with environmental sustainability. This is particularly relevant for Makindye Ssabagabo, where rapid urbanization demands integrated planning approaches that protect natural assets while supporting socio-economic development.

Table 1: Frameworks, Policies, Plans, and Acts relevant to the Environment in Makindye Ssabagabo Municipality

Frameworks, Policies, Plans and Acts	Relevance to Makindye Ssabagabo Municipality
International Frameworks	
The Ramsar Convention on Wetlands (1971)	This convention provides a framework for the conservation and wise use of wetlands. Its linkage to Makindye Ssabagabo is through Uganda's national commitments, as the municipality contains important wetland ecosystems, such as parts of the Lubigi wetland, parts of Lake Victoria's shores and associated ecosystem that are prone to encroachment and pollution from urban expansion. The convention supports guides municipal actions to protect these wetlands, preventing degradation from waste dumping and construction, which supports flood control, biodiversity conservation, and water purification—key for the municipality's growing population and vulnerability to flooding.
Basel Convention (1989)	This convention regulates transboundary movements of hazardous waste and promotes its environmentally sound management. Through Uganda's ratification, following increasing e-waste and

	hazardous waste in especially urban areas this convention encourages environmentally sound waste management practices in Makindye Ssabagabo, especially in informal settlements and industrial zones. It protects public health and the environment from hazardous waste pollution, such as from informal dumping sites in the municipality, by encouraging recycling and safe disposal practices to minimize soil and water contamination.
UN Framework Convention on Climate Change (UNFCCC) (1992)	The UNFCCC aims to stabilize greenhouse gas concentrations and foster international cooperation on climate mitigation and adaptation. It links to Makindye Ssabagabo through Uganda's national climate strategies, which require local governments to integrate climate actions. Makindye Ssabagabo, being part of the broader Kampala Metropolitan Area, contributes to and is affected by climate change. The municipality can implement mitigation (e.g., green building codes) and adaptation (e.g., flood drainage) measures in alignment with Uganda's national commitments. The UNFCC therefore, supports municipal efforts to address urban heat islands, flooding, and emissions from transport and waste, promoting resilience in a densely populated area prone to climate-induced disasters.
Convention on Biological Diversity (CBD) (1992)	The CBD focuses on conserving biodiversity, sustainable use, and fair sharing of genetic resources, of which Uganda is committed to. Uganda's biodiversity commitments are applicable to Makindye Ssabagabo's ecosystems like wetlands and green spaces threatened by urbanization. Urban expansion threatens natural habitats. This convention, therefore, guides municipal conservation efforts to protect local species and habitats, ensuring sustainable urban planning that balances development with ecosystem preservation.
UN Convention to Combat Desertification (UNCCD) (1994)	The UNCCD addresses desertification and land degradation through sustainable land management. Linked to the municipality through national anti-degradation strategies. Though desertification is less of a direct concern, land degradation from poor land use, poor farming on peri-urban lands, and deforestation is an issue in Makindye Ssabagabo Municipality. UNCCD Supports sustainable land management practices for combating soil erosion and land loss in Makindye Ssabagabo's hilly and urbanizing landscapes, supporting restoration projects to maintain agricultural productivity and prevent landslides in peri-urban areas.
Paris Agreement (2015)	The Paris Agreement limits global warming to well below 2°C and pursues efforts to limit it to 1.5°C. It promotes collective action on climate change and encourages countries to transition to low-carbon economies. This agreement aims to limit global warming and promote low-carbon economies. Its linkage to Makindye Ssabagabo is through Uganda's Nationally Determined Contributions (NDCs), requiring local implementation. It encourages

	municipal initiatives like green infrastructure and renewable energy adoption to reduce emissions from urban activities, enhancing climate resilience in a municipality facing rising temperatures and extreme weather.
Sustainable Development Goals (2015)	The SDGs address issues like climate change, biodiversity, and resource management. Linked to Makindye Ssabagabo Municipality through Uganda's national SDG integration, they are relevant to Makindye Ssabagabo by guiding local planning for goals such as SDG 11 (sustainable cities), SDG 13 (climate action), and SDG 15 (life on land), promoting balanced urban growth that mitigates environmental degradation in a rapidly expanding municipality. Makindye Ssabagabo Municipality's alignment with these goals will ensure sustainability.
United Nations Environment Programme (UNEP)	UNEP provides global guidance on environmental issues like climate change and pollution. Its linkage is through Uganda's participation in UNEP initiatives, influencing municipal policies. It supports local efforts in pollution control and biodiversity conservation and provides technical guidance and tools which the municipality can adopt to improve environmental governance, such as waste management programs to reduce urban pollution in Makindye Ssabagabo.
UN 2030 Agenda (2015)	This agenda emphasizes environmental protection and SDGs, including targets on climate and life below water. Linked to the municipality via national adoption, it supports integrated planning approaches in Makindye Ssabagabo that balance socio-economic growth with environmental sustainability. The agenda is relevant for integrating environmental goals into local development plans, such as reducing waste pollution in water bodies and promoting sustainable urban practices to ensure a healthy environment.
Regional Planning Frameworks	
African Union Climate Change Strategy (2022)	This strategy supports climate-resilient development and regional cooperation. Its linkage to Makindye Ssabagabo is through Uganda's alignment with African Union policies, informing local climate actions. Provides context for regional collaboration and funding mechanisms that Makindye Ssabagabo can leverage for climate-resilient infrastructure and planning. Its relevance further includes enhancing municipal adaptation to climate impacts like flooding, through knowledge sharing and strategies for urban resilience.
East African Climate Change Policy (2011)	This policy guides climate change adaptation and mitigation in East Africa, promoting regional cooperation and climate-resilient development. It supports countries in developing effective climate change response strategies. This policy guides regional climate adaptation and mitigation. Linked via East African Community frameworks, it is relevant to Makindye Ssabagabo by supporting local strategies for climate-resilient

	development, such as managing urban flooding and promoting green spaces in a region vulnerable to climate variability.
Eastern African Regional Framework Agreement on Air Pollution (2008)	Focuses on mitigating the adverse impacts of air pollution from various sources including waste disposal and incineration. By promoting cleaner practices, and reducing emissions, it supports sustainable waste management, minimizing harm to human health and the environment. This focuses on mitigating air pollution from sources like waste incineration. Its linkage is through regional commitments adopted nationally, applicable to urban air quality issues in Makindye Ssabagabo. Relevance lies in promoting cleaner waste practices to reduce emissions from open burning and industrial activities, protecting public health in a polluted urban setting.
African Convention on the Conservation of Nature and Natural Resources (1968, Revised 2017)	The convention promotes sustainable management and conservation of natural resources, and aims to protect the environment from pollution and degradation but indirectly supports waste management practices that minimize harm to ecosystems and human health by emphasizing the importance of environmental protection, thereby contributing to a more sustainable approach to waste management in Africa. This promotes sustainable management of natural resources and pollution prevention. Linked to the municipality via national implementation, it is relevant for supporting waste management that minimizes ecosystem harm, emphasizing environmental protection in Makindye Ssabagabo's urban and peri-urban areas.
National Planning Frameworks	
Uganda Vision (2040)	Uganda Vision 2040 aims to transform Uganda into a modern, prosperous nation while promoting sustainable development and environmental conservation. The vision emphasizes the need for green growth, sustainable natural resource management, and climate resilience to ensure a healthy environment and sustainable development. By 2040, the vision seeks to achieve a balanced and sustainable development that enhances the country's natural capital and mitigates environmental degradation. This vision promotes sustainable development and environmental conservation. Its direct linkage to Makindye Ssabagabo is as a national blueprint guiding municipal planning. Relevance includes fostering green growth and climate resilience, addressing local issues like deforestation and pollution to achieve balanced urban development by 2040.
Fourth National Development Plan (NDP IV) 2025/26-2029/30	The Fourth National Development Plan (NDP IV) prioritizes sustainable environmental management. Linked through national directives to local governments, it is relevant for Makindye Ssabagabo by promoting green growth, pollution control, and climate strategies to balance economic expansion with environmental protection in an urbanizing area. .

National Policies and guidelines	
The 1995 Constitution of Uganda (1995)	Articles 237(2) and 245 mandate environmental protection and sustainable management. Its linkage is constitutional, binding Makindye Ssabagabo as a local authority. Relevance is in integrating environmental concerns into municipal planning, preventing pollution and promoting awareness to ensure a clean environment.
The National Environment Management Policy (1994)	Provides a framework for environmental conservation in Uganda, promoting sustainable development and responsible use of natural resources. The policy plays a crucial role in protecting Uganda's environment, enhancing livelihoods, and ensuring a healthy and sustainable future for its citizens. the second objective requires entities to integrate environmental concerns in all development-oriented policies, planning and activities at national, district and local levels, with participation of the people. This provides a framework for environmental conservation and integration into development. Linked via national policy, it is relevant for Makindye Ssabagabo by requiring entities to incorporate environmental concerns in local planning, enhancing sustainable resource use in urban contexts.
National Wetlands Policy (1995)	This policy conserves and manages wetlands sustainably. Its linkage to the municipality is direct, given local wetlands. Makindye Ssabagabo Municipality is mandated in protecting ecosystems from urban encroachment, supporting wise use for flood mitigation and biodiversity.
The National Water Policy (1999)	This emphasizes water resource protection from pollution. Linked nationally, it is relevant for Makindye Ssabagabo by promoting waste management to prevent water contamination, ensuring sustainable water supply in an urban area.
Electronic Waste (E-Waste) Management Policy for Uganda (2012)	This framework manages e-waste through recycling and safe disposal. Its linkage is through national adoption, relevant to Makindye Ssabagabo's urban waste challenges, mitigating health and environmental risks from improper e-waste handling. This policy provides a framework for managing electronic waste, promoting environmentally sound disposal, and encouraging recycling. It aims to mitigate environmental and health risks associated with e-waste, while promoting sustainable practices in the electronics sector. It also promotes the implementation of regulations and standards for the management of e-waste, encourages the adoption of environmentally friendly technologies, and fosters public awareness and participation. By setting out clear objectives and actions, the policy seeks to improve the handling of electronic waste, reduce its adverse impacts, and support sustainable development goals while safeguarding public health and the environment in Uganda.
National Climate Change Policy (2015)	This supports low-carbon transitions and resilience. Linked to local implementation, it is relevant for addressing climate impacts like

	flooding in Makindye Ssabagabo through municipal adaptation measures.
Guidelines for E-Waste Management in Uganda (2016)	These provide standards for e-waste handling. Their linkage is national. The policy is relevant for reducing risks in Makindye Ssabagabo by promoting sustainable disposal practices in an area with growing electronic waste.
National Acts	
The Water Act (1995)	Water Act Cap 152. This regulates water use, protection and management and prohibits pollution. It guides municipal water management and is relevant for forming local associations to sustainably manage water resources and prevent waste-related pollution.
National Water and Sewerage Corporation Act (1995)	This Act manages water and sewerage services sustainably. Its linkage is through service provision in urban areas like Makindye Ssabagabo, relevant for minimizing environmental harm from sewerage and promoting public health.
The Local Governments Act, 1997	Cap 243, Schedule 2, Part II-V lays out that management of the Environment and Natural Resources (ENR) as mostly a decentralized service and points out key areas decentralized to local governments and urban councils. This decentralizes environmental management to local levels. Directly linked as it empowers Makindye Ssabagabo to handle waste, wetlands, and pollution, its relevance is in enabling local enforcement of environmental services.
The Land Act (1998)	This governs land use and promotes sustainability. Linked to municipal land planning, it is relevant for regulating urban land to prevent degradation in Makindye Ssabagabo.
National Forestry and Tree Planting Act (2003):	This act promotes sustainable forest management and conservation, regulating forest resource use and protection. It supports the country's efforts to conserve and restore forest ecosystems. This promotes forest conservation. Its linkage is national, relevant for restoring urban green spaces in the municipality to combat deforestation.
National Environment Act (2019):	This act regulates environmental management and conservation in Uganda, protecting natural resources and promoting sustainable development. It provides a framework for environmental governance and management. Act 5 2019. Section 3(1) suggests that every person in Uganda has a right to a clean and healthy environment in accordance with the Constitution and the principles of sustainable development. Section 3(2), Every person has a duty to create, maintain and enhance the environment, including the duty to prevent pollution. Cap 153 Section 18 (i) of the Act requires every district to prepare a District Environment Action plan to be revised every 3 years. This regulates environmental management, mandating rights to a clean environment and district action plans. Linked directly, it

	requires Makindye Ssabagabo to prepare and revise environmental plans, promoting pollution prevention.
The Uganda Wildlife Act (2019)	This act aims to conserve wildlife and their habitats in Uganda. Its effectiveness has been significant in protecting endangered species, but enforcement is crucial to prevent habitat destruction and poaching. The major purpose of this Act is to promote the conservation of wildlife throughout Uganda so that the abundance and diversity of their species are maintained at optimum levels commensurate with other forms of land use. This conserves wildlife and habitats. Its linkage is through national protection, relevant for preserving local biodiversity in Makindye Ssabagabo's ecosystems.
The Public Health Act	The Act plays a crucial role in protecting the environment and public health by regulating sanitation, waste management, and pollution control. The Act empowers authorities to take measures to prevent the spread of diseases and promote a clean environment, thereby contributing to environmental health and wellbeing. By enforcing provisions related to environmental sanitation and hygiene, the Act helps mitigate environmental health risks and promotes a healthier environment for Ugandans. This regulates sanitation and waste to protect health. Linked nationally, it is relevant for enforcing waste management in the municipality to prevent disease and pollution.
The Occupational Safety and Health Act (2006)	This protects workers and reduces hazards. Its linkage is indirect via workplace regulations, relevant for minimizing environmental risks from industrial activities in Makindye Ssabagabo.
National regulations	
The Petroleum (Exploration and Production) (Conduct of Exploration Operations) Regulations (1993).	It provides guidelines on ensuring environmental management in the petroleum sector with respect to prevention and control of pollution, ensuring health and safety during petroleum activities These ensure environmental management in petroleum activities. Linked nationally, it is relevant for pollution control in any local petroleum-related operations affecting Makindye Ssabagabo.
The National Environmental Impact Assessment (EIA) Regulations (1998)	The regulations offer guidelines on the minimization of negative environmental impacts from projects through aspects like project briefs and EIA reports. Further stipulations on the, approval or rejection of projects or cancellation of EIA certificates are provided. These minimize project impacts through assessments. Similarly, Makindye Ssabagabo municipality refers to the EIA regulations for local project approvals in the municipality as they are relevant for ensuring sustainable development in the municipality.
The National Environment Waste Management Regulation (1998)	This covers waste handling and the 3Rs. Directly linked, relevant for comprehensive waste management in urban Makindye Ssabagabo. Encompasses all types of hazardous and non-hazardous waste along the value chain including storage, dumping, transportation, disposal, reducing, reuse and recovery (3Rs).

The National environment (Wetlands, River Banks and Lake Shores Management) Regulations (2000)	Provides guidelines for the conservation and wise use of wetlands, riverbanks and Lakeshores and the resources therein and in line with the public trust dogma clause 2 of article 237 of the Constitution (public trust doctrine). These guide wetland conservation and local enforcement, relevant for protecting Makindye Ssabagabo's water-adjacent areas.
The National Environment (Hilly and Mountainous Areas Management) Regulations (2000)	The regulations aim to conserve and protect Uganda's hilly and mountainous areas from degradation, promoting sustainable land use and environmental protection. They help mitigate environmental risks such as soil erosion, landslides, and loss of biodiversity in these ecologically sensitive areas. Therefore, these regulations are relevant for erosion control in Makindye Ssabagabo's terrain.
National Environment (Standards for Discharge of Effluent into Water or Land) Regulations (2020)	These regulations set standards for effluent discharge to prevent water pollution. Their effectiveness relies on regular monitoring and enforcement to ensure compliance. Similarly, for Makindye Ssabagabo, these regulations provide guidance on initiatives for preventing water pollution from municipal sources.

Planning Process and Methodology

The preparation of the Makindye Ssabagabo Environmental Action Plan followed a participatory, evidence-based, and results-oriented approach. The process was anchored on the findings of the State of the Environment Report, which provided the baseline data and trend analysis for key environmental themes.

1. **Inception Phase** – The process began with an inception meeting at the municipal headquarters, where the scope, objectives, and timelines for the EAP were agreed upon. This phase also mapped key stakeholders, including municipal departments, division administrations, community representatives, civil society organizations (CSOs), private sector actors, and relevant national agencies such as NEMA and the Ministry of Water and Environment.
2. **Data Collection and Situation Analysis** – Data was gathered from multiple sources, including municipal records, thematic reports, GIS mapping, and on-site inspections. Field assessments were conducted to capture the state of wetlands, riverbanks, waste management facilities, biodiversity areas, and pollution hotspots. Key Informant Interviews (KIIs) and Focus Group Discussions (FGDs) were held with community leaders, local councils, school representatives, market vendors, and youth groups to capture local perspectives.

3. **Stakeholder Consultations** – Structured consultations were conducted in all divisions to validate findings and identify priority issues. These sessions also served to generate consensus on feasible interventions, taking into account available resources and institutional capacities.
4. **Results-Based Planning** – The issues identified were translated into result areas, outcomes, outputs, and specific actions. Priority locations, timeframes, and responsible entities were assigned based on technical feasibility and impact potential.
5. **Review and Validation** – A draft plan was reviewed by the Municipal Technical Natural Resources Management Committee to ensure alignment with the Municipal Development Plan and other sectoral strategies.

2. STRATEGIC FRAMEWORK

2.1 Vision, Mission, and Goal of the EAP

Vision

A clean, green, climate-resilient Makindye Ssabagabo where environmental sustainability drives socio-economic transformation.

Mission

To promote sustainable environmental management through coordinated action, community participation, innovation, and effective enforcement of policies for the well-being of present and future generations.

Goal

To restore, protect, and sustainably manage the environmental resources of Makindye Ssabagabo Municipality, ensuring improved ecosystem health, enhanced resilience to climate change, and better quality of life for all residents.

2.2 Guiding Principles

i. Sustainability – Ensuring that all interventions balance ecological health, social welfare, and economic viability.

ii. Evidence-Based Decision-Making – Basing priorities and actions on accurate environmental data and scientific assessments.

iii. Inclusivity and Participation – Involving communities, CSOs, private sector actors, and marginalized groups in environmental planning and implementation.

iv. Transparency and Accountability – Promoting openness in resource allocation, implementation, and reporting.

v. Partnerships and Collaboration – Working with local, national, and international partners for technical and financial support.

vi. Climate Resilience – Mainstreaming climate adaptation and mitigation in all environmental actions.

vii. Polluter Pays Principle – Holding individuals and institutions accountable for environmental degradation

2.3 Alignment with National and Municipal Development Priorities

The Makindye Ssabagabo Environmental Action Plan aligns with the Uganda Vision 2040 which calls for sustainable natural resource management to support economic transformation; the National Development Plan III (NDP III & IV) – particularly the Programme on Natural Resources, Environment, Climate Change, Land, and Water Management; National Environment Act, 2019 – which mandates local governments to safeguard natural resources and integrate environmental concerns into all development plans; Paris Agreement on Climate Change – by promoting climate adaptation, resilience, and low-carbon development; and Sustainable Development Goals (SDGs) – specifically SDG 6 (Clean Water and Sanitation), SDG 11 (Sustainable Cities), SDG 13 (Climate Action), SDG 14 (Life Below Water), and SDG 15 (Life on Land).

At the municipal level, the EAP directly supports the Makindye Ssabagabo Municipal Development Plan, which prioritizes improved urban management, increased green cover, enhanced waste management systems, and climate change adaptation. It also complements sectoral plans in health, tourism, and infrastructure, ensuring that environmental considerations are integrated into all aspects of local development.

2.4 Key Environmental Challenges

The State of the Environment Report for Makindye Ssabagabo highlights a number of interlinked environmental challenges that threaten the municipality's ecological integrity, public health, and long-term socio-economic stability.

Wetland and Riverbank Degradation – Unregulated settlement expansion, agricultural encroachment, and land reclamation have significantly reduced wetland and riparian zones, undermining their role in flood control, water filtration, and biodiversity support. Key wetlands such as Lufuka, Kaliddubi, and Kyetinda are particularly affected.

Water Pollution – Both surface and groundwater resources are under pressure from domestic wastewater, industrial effluents, and solid waste dumping. Field observations indicate reddish-brown and dark-grey water in several wetlands, sluggish flow, and stagnant ponding, all symptomatic of heavy organic load and siltation.

Poor Solid Waste Management – Rapid population growth and urbanization have outpaced waste collection capacity. Informal settlements experience irregular or no waste collection, resulting in illegal dumping in drainage channels, wetlands, and open spaces.

Loss of Biodiversity and Green Cover – Conversion of forests, grasslands, and wetland habitats to residential and commercial use has led to a decline in native species. Tree cover loss is accelerating, particularly in hillside areas such as Bunamwaya and peri-urban corridors.

Air and Noise Pollution – Increasing motorized transport, boda-boda operations, and small-scale industries contribute to deteriorating air quality. Noise pollution is also rising in commercial centres and transport hubs.

Climate Change Vulnerability – The municipality is increasingly prone to flooding, especially in low-lying areas such as Ndejje, due to reduced natural drainage capacity, intense rainfall events, and unplanned construction in floodplains.

Weak Enforcement and Governance – Environmental regulations are inconsistently enforced due to inadequate staffing, lack of monitoring equipment, and occasional conflicts between local authorities and national agencies such as NEMA.

Low Public Awareness and Participation – Many residents still view wetlands as wastelands or idle land for farming and construction, and there is limited knowledge of environmental laws, sustainable practices, and the long-term benefits of conservation.

These challenges underscore the urgent need for coordinated, adequately resourced interventions—precisely the role of this EAP. The EAP directly responds to these issues by prioritizing restoration of degraded ecosystems, strengthening enforcement, improving waste and pollution management, enhancing climate resilience, and promoting community-led environmental stewardship.

2.5 Crosse-cutting Issues

The Environmental Action Plan for Makindye Ssabagabo integrates several cross-cutting themes to ensure that environmental interventions are inclusive, equitable, and sustainable. These themes—Gender, Youth, Equity, and Climate Resilience—are mainstreamed across all result areas, outputs, and actions.

Gender

Women play a central role in natural resource management, particularly in water collection, household waste disposal, and small-scale agriculture. However, they often face barriers to participating in decision-making processes and accessing environmental resources. The EAP

promotes gender-responsive planning and budgeting to ensure women's needs and perspectives are fully addressed. This includes:

- Encouraging women's participation in environmental governance committees and user associations.
- Supporting women-led enterprises in recycling, eco-tourism, and climate-smart agriculture.
- Tailoring environmental awareness campaigns to address gender-specific roles and challenges.

Youth

Makindye Ssabagabo has a predominantly young population, many of whom are unemployed yet possess the energy and creativity needed for environmental transformation. The EAP incorporates targeted youth engagement by:

- Creating green jobs in waste management, tree planting, and wetland restoration.
- Establishing school-based eco-clubs and municipal youth environmental awards.
- Promoting vocational training in renewable energy, climate-smart farming, and green enterprise development.

Equity

Equity considerations ensure that vulnerable groups—including persons with disabilities, the elderly, and the urban poor—benefit equally from environmental interventions. In the EAP:

Waste collection and water resource protection activities will target underserved informal settlements.

Capacity-building programs will include provisions for accessibility and inclusivity.

Livelihood initiatives will prioritize marginalized households dependent on fragile ecosystems for income.

Climate Resilience

Climate change impacts—particularly flooding and prolonged dry spells—pose significant risks to the municipality's environment, economy, and public health. The EAP integrates climate resilience in all interventions by:

- Restoring wetlands and riverbanks to enhance natural flood control.

- Mainstreaming climate risk assessments in land-use planning and infrastructure development.
- Promoting household and community adaptation measures such as rainwater harvesting, flood early warning systems, and the use of climate-resilient crop varieties.

These cross-cutting issues will not be treated as stand-alone components but will be embedded in the design, implementation, and monitoring of every EAP activity. This ensures that environmental gains are socially inclusive, economically beneficial, and resilient to climate shocks.

3. CHAPTER THREE: RESULT-BASED PLANNING

3.1 Overview of the Results Framework

The Makindye Ssabagabo Environmental Action Plan is anchored on a results-based framework, a structured planning, communication, and management tool designed to focus interventions on measurable outcomes. The results framework concept, popularized in the mid-1990s by USAID, has since become a standard in development planning, providing clarity on objectives, logical linkages between activities, and measurable performance indicators.

A results framework, much like a log frame or theory of change, provides a clear pathway from inputs and activities to outputs, outcomes, and long-term impacts. It allows practitioners to visualize the cause-and-effect relationships underlying interventions, identify performance indicators at each stage, and anticipate potential risks to achieving objectives.

For the Makindye Ssabagabo EAP, the results framework serves multiple purposes including planning to ensure that environmental interventions are logically sequenced, targeted, and achievable within the available timeframe and resources; management to enable the Municipal Council and partners to track progress, adjust strategies, and allocate resources effectively and communication by way of providing stakeholders with a clear, visual representation of how the EAP's actions contribute to broader environmental goals.

The framework is built on sound logic, technical analysis, established environmental management principles, and the contextual expertise of local stakeholders. Each result area—whether in waste management, water resource protection, biodiversity, pollution control, climate resilience, or community capacity building—is linked to specific outputs, actions, and measurable indicators. This ensures that implementation is both impact-focused and accountable, while maintaining flexibility to respond to emerging environmental challenges.

3.2 Result Areas & Outcomes

3.2.1 Waste Management

Makindye Ssabagabo faces significant solid waste challenges, with illegal dumping in markets, informal settlements, and along drainage channels. Waste collection coverage is low, and recycling remains minimal. Poor disposal practices lead to blocked drainage, flooding, and pollution of wetlands and water bodies. Action is needed to enforce waste ordinances, expand recovery and composting facilities, introduce economic incentives for waste segregation, and promote community-led recycling campaigns to align with SDGs.

Outcome 1: Waste management is improved

Output 1: Waste management improved

Key Actions/Interventions

The proposed actions to address environmental challenges related to waste include

Short-term (0–2 yrs)

1. Enforce penalties for illegal dumping - Strengthen and implement municipal ordinances to deter indiscriminate disposal of waste in markets, informal settlements, and drainage channels, ensuring regular enforcement patrols.
2. Establish waste material recovery facilities - Set up designated centers for sorting, recycling, and reusing materials, working with private waste companies and community cooperatives.
3. Distribute household waste separation kits -

Medium-term (2–5 yrs)

1. Introduce green fees/environmental levies on markets and businesses; Implement charges for markets, businesses, and large events to generate revenue for waste management services and environmental projects
2. Deploy communal waste collection points - Establish strategically located waste collection centers in high-density residential areas, trading centers, and peri-urban settlements to provide residents with accessible and safe waste disposal facilities
3. Launch recycling awareness drives - conduct rotating outreach campaigns across parishes to encourage households, schools, and businesses to segregate and recycle waste.
4. Develop municipal composting sites - Construct and operate composting facilities in peri-urban areas to process organic waste from markets and households into usable manure.

Long-term (5+ yrs)

1. Upgrade waste collection fleet - Replace or supplement the municipality's existing waste collection vehicles with modern, higher-capacity, and fuel-efficient trucks that are better suited for both formal and informal settlement areas
2. Integrate circular economy principles - Shift waste management from a linear "collect–dispose" model to a circular approach that prioritizes waste reduction, reuse, recycling, and recovery of materials.

3.2.2 Water Resource Protection

The municipality's wetlands and drainage systems such as Kaliddubi, Lufuka, and Kyentinde are under pressure from encroachment, siltation, and wastewater discharge. The SoE showed that about 60% of the wetlands and rivers have been lost since 2005. The water sources are at risk from pollution by car washing bays, industries, and poor sanitation. Wetlands provide flood control, water purification, and biodiversity support. Their protection is critical for climate resilience and compliance with the National Environment Act (2019). Mapping, demarcation, and strict bylaw enforcement are urgent, alongside promoting eco-tourism, community stewardship, and rainwater harvesting to safeguard these vital ecosystems.

Outcome 2: Water resources are protected

Output 2: Water resource protection enhanced

Key Actions/Interventions

Therefore, the proposed management actions include:

Short-term

1. Demarcate wetland boundaries using GIS and field surveys to establish clear, legally recognized wetland boundaries, installing visible markers.
2. Identify/map critical water sources and drainage channels - Survey and map springs, boreholes, and key drainage channels to prioritize protection and restoration activities.
3. Map water sources/drainage channels.

Medium-term

1. Enforce bylaws against encroachment on wetlands and river banks by deploying enforcement teams to monitor wetland areas and take action against illegal activities such as construction and dumping.
2. Promote sustainable wetland use through eco-tourism, controlled harvesting, and community-led conservation programs to balance livelihoods with ecosystem protection.
3. Promote rainwater harvesting and community water conservation practices by facilitating adoption of rooftop collection systems and community tanks in water-stressed areas to reduce reliance on polluted surface sources.
4. Regulate polluting activities (e.g., car washing bays) by licensing and monitoring car wash operations to prevent wastewater discharge into rivers, wetlands, and drainage systems.

Long-term

1. Establish community-led wetland management groups
2. promote eco-tourism initiatives.

3.2.3 Urban Greening & Biodiversity Protection

Urbanization has led to rapid loss of tree cover, green corridors, and biodiversity-rich habitats in Makindye Ssabagabo. Limited tree cover exacerbates heat islands and reduces wildlife habitats. Urban forests mitigate climate impacts, enhance air quality, and support biodiversity. Green spaces improve livability and align with Uganda's Forest Act. Therefore, this result area is aimed at increase forest cover and green space around the Municipality. Degraded hillsides, wetland buffers, and community spaces require restoration through large-scale tree planting, community woodlots, and agroforestry. Biodiversity hotspots must be mapped and protected, with awareness campaigns to curb illegal hunting and habitat destruction, and initiatives to promote urban biodiversity for both ecological and recreational value.

Outcome 3: Urban green spaces are increased and biodiversity is protected

Outputs: Urban greening increased and biodiversity protected

Key Actions/Interventions

The actions to increase forest cover and green spaces around the municipality include;

Short-term

1. Conduct tree cover assessments in the municipality
2. Launch municipal tree-planting campaigns - Coordinate seasonal planting drives in schools, roadsides, and community open spaces, engaging local residents and corporate sponsors
3. Map biodiversity hotspots - Identify and document ecologically rich sites such as wetlands, forest patches, and green corridors to guide conservation priorities.

Medium-term

1. Launch annual tree-planting campaigns
2. Develop community woodlots and promote agroforestry practices - Support establishment of small-scale communal forests for firewood and timber, reducing pressure on natural ecosystems.
3. Raise awareness/enforce laws against illegal hunting and habitat destruction - Educate communities about biodiversity laws while strengthening enforcement against poaching, illegal logging, and habitat destruction.

Long-term

1. Launch annual tree-planting campaigns
2. Develop community woodlots and promote agroforestry practices
3. Raise awareness/enforce laws against illegal hunting and habitat destruction

3.2.4 Pollution Control

Air, water, and land pollution threaten community health and the environment. Air pollution from vehicle and boda-boda emissions, industrial effluents, open burning of waste, and poor fuel quality are a major environmental challenge in Makindye Ssabagabo Municipality. Pollution causes respiratory diseases, ecosystem toxicity, and violates SDG 3 (Health). Monitoring and regulation are vital for public/environmental health. Establishing air quality monitoring stations, enforcing emission regulations, and promoting cleaner transport options are essential to curb pollutants. Public awareness campaigns will complement enforcement to foster behavior change.

Outcome: Pollution of all kinds is controlled

Output 4: All kinds of pollution controlled

Key Actions/Interventions

Short-term

1. Install air quality monitoring stations in urban centers - Deploy sensors in high-traffic and industrial zones to monitor pollutants and guide policy actions.

Medium-term

1. Regulate emissions from boda-bodas, vehicles, and industries - Enforce standards for vehicle, motorcycle (boda-boda), and industrial emissions to improve air quality.
2. Ban waste burning - Control burning of waste in urban and peri-urban areas while providing safe waste disposal alternatives and public education.
3. Promote use of cleaner fuels and non-motorized transport (bicycles) - Encourage transition to LPG, solar, and electric vehicles, and invest in cycling and pedestrian infrastructure.

Long-term

1. Expand non-motorized transport (cycling lanes)
2. Green industrial certifications.

3.2.5 Climate Resilience

Makindye Ssabagabo is highly vulnerable to climate-related hazards such as floods, prolonged dry spells, and fires, especially in informal settlements and low-lying areas like Ndejje. Climate-smart agriculture, early warning systems, and disaster preparedness measures are urgently required. Developing a municipal climate adaptation strategy, promoting energy-efficient technologies, and strengthening disaster risk management will enhance community resilience, aligning with Uganda's NDC under the Paris Agreement.

Outcome 5: Climate resilience is enhanced

Output 5: Climate resilience enhanced

Key Actions/Interventions

The proposed actions to realize increased climate resilience in the municipality include the following.

Short-term

1. Develop a climate change adaptation/mitigation strategy outlining adaptation priorities in water, agriculture, infrastructure, and disaster risk reduction.
2. Update disaster risk management plan
3. Conduct disaster preparedness drills - Organize regular simulation exercises to improve readiness among residents, schools, and institutions.
4. Build community-based disaster response committees - Form local teams trained in first response, evacuation coordination, and resource distribution.
5.

Medium-term

1. Promote energy-efficient technologies - Encourage adoption of clean cookstoves, solar lighting, and energy-saving appliances for households and businesses
2. Establish early warning systems (for floods/drought)
3. Integrate climate-smart agriculture - Support farming practices that conserve water, build soil health, and increase resilience to climate variability.
4. Establish flood early-warning systems. Install flood and fire alert systems linked to community notification networks.

Long-term

1. Restore ecosystems for natural buffering; Build climate-proof infrastructure.

2. Develop a post-disaster recovery plan outlining measures for the rapid restoration of services, livelihoods, and infrastructure following disasters.

3.2.6 Community Awareness & Capacity Building

Limited technical capacity, weak coordination, and low public awareness hinder environmental management in Makindye Ssabagabo. There is a shortage of trained environmental officers, and governance structures need strengthening. Community awareness campaigns, formation of environmental watch committees, and integration of environmental focal points in divisions will enhance ownership and compliance. Partnerships with CSOs, the private sector, and schools are key to sustaining progress. This result area aligns with SDG 17 (Partnerships).

Outcome 6: Community awareness and capacity building are increased

Output 6: Community awareness and capacity building enhanced

Key Actions/Interventions

Short-term

1. Conduct an institutional audit to assess the current capacity, staffing, structures, and processes for environmental governance to identify gaps and improvement areas.
2. Create an environment management department - Establish a dedicated municipal unit to coordinate, implement, and monitor environmental activities.
3. Integrate environmental focal points in divisions/wards - Assign specific officers in each division/ward to serve as liaison points for environmental matters.
4. Recruit environmental officers at municipal/division levels - Hire qualified professionals at municipal and division levels to strengthen enforcement, monitoring, and planning.
5. Establish an environmental governance committee

Medium-term

1. Conduct in-service training - Provide ongoing professional development for municipal staff and local leaders on topics such as environmental assessments, climate change, and policy implementation.
2. Develop career progression paths for environmental staff - Create structured promotion and growth opportunities for environmental officers to retain talent and encourage specialization.
3. Establish Environmental Governance Committee - Form a multi-stakeholder body to provide oversight, coordination, and strategic guidance on environmental issues.

4. Form Community Environmental Watch Committees - Mobilize and train local volunteers to monitor environmental violations, report incidents, and lead awareness efforts.
5. Develop partnerships with CSOs, youth, and the private sector - Collaborate with civil society organizations, private sector actors, and youth groups to expand resources and reach.
6. Create dedicated Environment Fund - Set aside municipal budget lines for environmental initiatives, supplemented by donor or private sector contributions.
7. Mobilize external funding - Pursue grants and technical support from development partners such as the World Bank, UNDP, and GIZ.
8. Review/harmonize bylaws with the National Environment Act & SDGs

Long-term

1. Develop Local Environment Management Policy - Draft and adopt a comprehensive local policy to guide long-term environmental management priorities
2. Create citizen scorecards for accountability.

4. IMPLIMENTATION PLAN

4.1 Introduction

The implementation framework for the Makindye Ssabagabo EAP (Table 2) provides a roadmap for executing key environmental interventions in the municipality from 2026 to 2030. Grounded in field assessments and community feedback, the framework organizes actions into six strategic result areas: Waste Management, Water Resource Protection, Urban Greening and Biodiversity, Pollution Control, Climate Resilience, and Community Awareness and Capacity Building. Each action is aligned with specific outputs and outcomes, supported by designated responsible institutions, timeframes (short, medium, or long-term- S, M, L), and targeted geographical locations across the municipality

Table 2: Implementation Plan for the Environmental Actions

Results Area/Outcome	Outputs	Actions	Priority locations	Timeframe	Responsible Entity
1. Waste Management Waste management is improved	Waste management improved	1. Enforce penalties for illegal dumping through strengthened ordinances	Markets, informal settlements	S - L	Law Enforcement & Environment Office
		2. Establish waste material recovery facilities	Central, Ndejje Trading Centre, Nde Market	S	Municipal Council and Private Waste Companies (e.g., Nabugabo Upcycle)
		3. Develop municipal composting sites	Peri-urban areas (e.g. Bunamwaya)	M	Agricultural Office and Youth Groups
		4. Introduce green fees/environmental levies on markets and businesses	Commercial zones, markets	M	Legal & Finance Units
		5. Organize Monthly Recycling Awareness Drives	All parishes (rotating locations)	S - L	Environment Officers and Division Health Inspectors
2. Water Resource Protection Water resources are protected	Water resource protection enhanced	1. Map and demarcate all wetland boundaries	Kaliddubi, Lufuka, Kyetinda wetlands	S	Environment Department & NEMA
		2. Enforce bylaws against wetland encroachment and illegal dumping	Encroached wetlands - Lufuka	S - L	Law Enforcement & Environment Officers
		3. Promote sustainable wetland use through eco-tourism and community management groups	Restorable wetlands	M	Tourism Office & NGOs
		4. Identify/map critical water sources and drainage channels	Lake Victoria shores, drainage basins	S	Water Department & Planners
		5. Regulate car washing bays and enforce penalties for water pollution	Landing sites, industrial zones - Busabala	S - L	Law Enforcement Unit

		6. Promote rainwater harvesting and community water conservation practices	Water-stressed areas (Ndejje, Bunamwaya)	M	Division Offices
3. Urban Greening & Biodiversity Urban green spaces are increased and biodiversity is protected	Urban greening increased and biodiversity protected	1. Conduct tree cover assessment and identify degraded areas	Hillsides (Bunamwaya, urban corridors	S	Forest Officer & GIS Unit
		2. Launch municipal tree-planting campaigns	Schools, roadsides, peri-urban areas	S/M	Schools, CSOs, Private Sector
		3. Develop community woodlots and promote agroforestry practices	Degraded lands, community spaces	M	Agricultural Extension Office
		4. Map biodiversity hotspots (wetlands, green corridors)	Wetlands, forests	S	Environment Dept. & NFA
		5. Raise awareness/enforce laws against illegal hunting and habitat destruction	Biodiversity-rich areas	S - L	Law Enforcement & CSOs
		6. Promote urban biodiversity (birdwatching zones, botanical gardens)	Green spaces, wetlands - Lufuka	L	Tourism & Parks Dept.
4. Pollution Control Pollution of all kinds is controlled	Pollution controlled	1. Install air quality monitoring stations	Urban centers – Ndejje,	M	Environment Department & NEMA
		2. Regulate emissions from boda-bodas, vehicles, and industries	High-traffic areas, industrial zones	S - L	Traffic Police, Transport Office
		3. Promote use of cleaner fuels and non-motorized transport (bicycles)	Municipality-wide	M	Planning & Engineering
		4. Control burning of waste and introduce awareness campaigns	Waste hotspots- Ndejje,	S - L	Environment Department, NEMA

5. Climate Resilience Climate resilience is enhanced	Climate resilience enhanced	1. Develop Municipal Climate Change Adaptation/Mitigation Strategy	Municipality-wide	S	Planning Unit & Ministry of Water & Environment
		2. Promote energy-efficient technologies (clean cook stoves, solar)	Informal settlements	M	Environment Dept. & NGOs
		3. Integrate climate-smart agriculture and flood early-warning systems	Flood-prone areas (Mutungo, Ndejje)	M	Agricultural Office
		4. Develop/update Municipal Disaster Risk Management Plan	Disaster-prone areas	S	Disaster Management Unit, Municipal Council
		5. Establish early warning systems for floods/fires	High-risk communities	M	Environment & IT Departments
		6. Conduct disaster preparedness drills	Vulnerable parishes	S - L	Disaster Management Unit
		7. Build community-based disaster response committees	Parish/division levels	S	Local Leaders & Environment Dept.
		8. Develop post-disaster recovery plan	Disaster-affected areas - Ndejje	L	Planning & Disaster Management Units
6. Community Awareness & Capacity Building Community awareness and capacity building are increased	Community awareness and capacity building enhanced	1. Conduct institutional audit for environmental governance	Municipality-wide	S	Municipal Council Admin Executive & MoLG
		2. Create Environment Management Department	Municipality-wide	S	Municipal Executive Committee
		3. Integrate environmental focal points in divisions/wards	All divisions/wards	S	Municipal Human Resource Office
		4. Recruit environmental officers at municipal/division levels	Municipality-wide	S	HR Department
		5. Conduct in-service training (environmental assessment, climate change)	Municipality-wide	M	Ministry of Water & Environment, NEMA, NGOs

	6. Develop career progression paths for environmental staff	Municipality-wide	M	HR & Public Service Commission
	7. Establish Environmental Governance Committee	Municipal HQ	S	Council Speaker's Office
	8. Form Community Environmental Watch Committees	All divisions/wards	S	Division Offices & CSOs
	9. Develop partnerships with CSOs, youth, and private sector	Municipality-wide	M	Environment Department
	10. Create dedicated Environment Fund	Municipality-wide	S	Finance Department
	11. Mobilize external funding (World Bank, UNDP, GIZ)	Municipality-wide	M	Planning Unit & Environment Department
	12. Review/harmonize bylaws with National Environment Act & SDGs	Municipality-wide	S	Legal Department & Environment Office
	13. Develop MSMC Local Environment Management Policy	Municipality-wide	M	Council, Legal & NEMA

4.2 Costing of Actions

The total projected budget (Table 3) for implementing the EAP over five years is approximately UGX 12 billion, spread across the six result areas. Budget needs vary depending on the scale and duration of activities. High-cost areas include infrastructure development (e.g., composting sites, air monitoring stations), training and staffing of environmental officers, biodiversity restoration, and climate resilience measures. Short-term priorities such as environmental bylaw enforcement, committee formation, and community awareness are relatively low-cost and highly impactful, offering good entry points for early implementation. The detailed breakdown of the cost per action under each result area can be found in Appendix 1:

Table 3: Summary of cost of Implementation

Result Area	2026	2027	2028	2029	2030	Total (UGX)
Waste Management	812	741	631	532	470	3186
Water Resource Protection	682	503	422	440	324	2371
Urban Greening & Biodiversity	485	351	394	257	232	1719
Pollution Control	304	253	212	154	121	1044
Climate Resilience	492	486	625	361	334	2298
Community Awareness & Capacity Building	425	355	331	254	265	1630
TOTAL	3200	2689	2615	1998	1746	12248

4.3 Funding Sources

Funding will be drawn from a combination of municipal own-source revenue, national government transfers, development partner grants, and public-private partnerships. This diversified financing approach will ensure financial sustainability and flexibility over the implementation period. Specific sources include:

i. Municipal green levies and environmental fines,

ii. National climate adaptation and environment budgets,

iii. NGO co-financing, especially for awareness, training, and livelihoods,

iv. Donor grants for infrastructure, disaster resilience, and governance reforms,

v. CSR investments from the private sector.

5. MONITORING, EVALUATION, AND LEARNING

5.1 Introduction

The Monitoring and Evaluation Framework (M&E) is a central component of the Makindye Ssabagabo Environmental Management Action Plan (Table 4). It provides the structure for tracking the effectiveness, efficiency, and impact of interventions under six key environmental result areas: waste management, water resource protection, urban greening and biodiversity, pollution control, climate resilience, and community awareness and capacity building. The M&E framework promotes accountability, supports adaptive management, and ensures alignment with national environmental policies and global sustainability targets by systematically monitoring outputs and outcomes.

5.2 Definition of Indicators

Indicators are essential tools for measuring the performance and success of environmental interventions. The EMAP uses three main types of indicators. Output indicators measure the direct results of specific actions, such as the number of composting sites developed or trees planted. Outcome indicators reflect broader changes that occur as a result of these actions, such as a reduction in illegal dumping or improved air quality. Impact indicators track long-term environmental improvements, such as increased biodiversity or reduced vulnerability to climate-related disasters. Each result area is associated with clearly defined indicators to guide progress monitoring and performance evaluation.

5.3 Data Collection and Analysis

Data collection will be carried out using a mix of tools and methods suitable for environmental monitoring. These include GIS mapping for wetland boundaries and green cover; structured surveys to measure changes in awareness, practices, and perceptions; focus group discussions for qualitative insights; and field inspections to assess infrastructure and compliance. Output-level data, such as the number of training sessions or recovery facilities, will be collected quarterly, while outcome and impact data—such as improved waste practices or increased climate resilience—will be assessed annually. Collected data will be analyzed using trend and gap analysis to identify progress, challenges, and opportunities for improvement across parishes and result areas.

5.4 Reporting and Feedback Mechanism

The EMAP features a tiered reporting structure to ensure regular feedback and inclusive participation. At the community level, quarterly feedback will be shared through barazas and community radio to keep residents informed and engaged. Technical reporting, including progress and monitoring reports, will also occur quarterly and will be submitted to key stakeholders like the Municipal Environment Department, NEMA, and the Ministry of Water and Environment. Strategic reporting will take place annually through performance review sessions targeting the Municipal Council, donors, and key partners. Additionally, the public will receive annual updates through media briefings and simplified visual reports, helping to foster transparency and public accountability.

5.5 Stakeholder Engagement

A strong and inclusive stakeholder engagement system underpins the M&E framework. The proposed Environmental Management Department will coordinate implementation, while environmental focal persons in divisions and wards will manage activities on the ground. The Environmental Governance Committee will provide oversight, and Community Environmental Watch Committees will support monitoring, reporting, and awareness at the grassroots level.

Capacity building will focus on training technical staff, community leaders, and youth in environmental governance, monitoring tools, and climate change adaptation. Equipment such as mapping kits, field forms, and mobile-based data tools will be provided to improve accuracy and participation. Partnerships will also be established with civil society organizations, the private sector, and academic institutions to enhance technical support and innovation.

Communication and advocacy efforts will include radio talk shows, school outreach, printed materials, drama performances, and social media campaigns in English and Luganda. These will promote behaviour change, share success stories, and build widespread support for environmental initiatives

Table 4: M&E framework for Makindye Ssabagabo Environment Action Plan

Result Area	Output	Indicator	Means of Verification	Baseline	2026	2027	2028	2029	2030
Waste Management	Waste management improved	No. of enforcement actions against illegal dumping	Municipal enforcement records	50	100	150	200	250	300
		No. of waste material recovery facilities established	Facility inspection reports	1	3	5	7	9	10
		No. of municipal composting sites developed	Project completion reports	0	1	2	3	4	5
		Revenue from green fees/environmental levies (UGX M)	Municipal revenue reports	0	20	40	60	80	100
		No. of monthly recycling awareness events held	Campaign reports	0	12	24	36	48	60
Water Resource Protection	Water resource protection enhanced	Km of wetlands demarcated	GIS maps, field verification	5	15	25	35	45	50
		No. of enforcement actions on wetland encroachment	Enforcement and legal records	10	30	50	70	90	110

		No. of eco-tourism/community wetland projects initiated	Project reports	1	3	5	7	9	10
		No. of water sources and drainage channels mapped	GIS and hydrological data	10	30	50	70	90	100
		No. of penalties enforced on car washing bays	Inspection and compliance reports	0	10	20	30	40	50
		No. of households adopting rainwater harvesting	Household surveys	50	100	200	300	400	500
Urban Greening & Biodiversity	Urban greening increased and biodiversity protected	Area (ha) of trees planted	Tree planting records	5	20	40	60	80	100
		No. of community woodlots established	Community forestry reports	2	4	6	8	10	12
		No. of biodiversity hotspots mapped	Biodiversity survey reports	3	6	9	12	15	18
		No. of awareness campaigns on wildlife protection	Event reports	0	5	10	15	20	25
		No. of urban biodiversity projects launched	Project reports	1	2	3	4	5	6

Pollution Control	Pollution controlled	No. of air quality monitoring stations installed	Installation reports	0	2	4	6	8	10
		No. of emission checks conducted	Inspection reports	100	300	500	700	900	1000
		No. of awareness events promoting clean fuels	Event reports	0	5	10	15	20	25
		No. of penalties issued for open waste burning	Enforcement records	10	30	50	70	90	110
Climate Resilience	Climate resilience is enhanced	Climate strategy developed (Yes/No)	Official strategy documents	0	1	1	1	1	1
		No. of energy-efficient technologies adopted	Household surveys	100	200	400	600	800	1000
		No. of disaster risk management plans updated	Municipal plans	0	1	1	1	1	1
		No. of early warning systems operational	EWS system logs	0	1	2	3	4	5
		No. of community disaster drills conducted	Drill reports	0	2	4	6	8	10
Community Awareness & Capacity Building	Community awareness and capacity	No. of institutional audits conducted	Audit reports	0	1	2	3	4	5

	building enhanced							
		No. of officers recruited/trained	HR/training records	5	10	15	20	25	30
		No. of community watch committees formed	Committee registers	0	5	10	15	20	25
		Amount mobilized from external sources (UGX M)	Financial reports	0	200	400	600	800	1000

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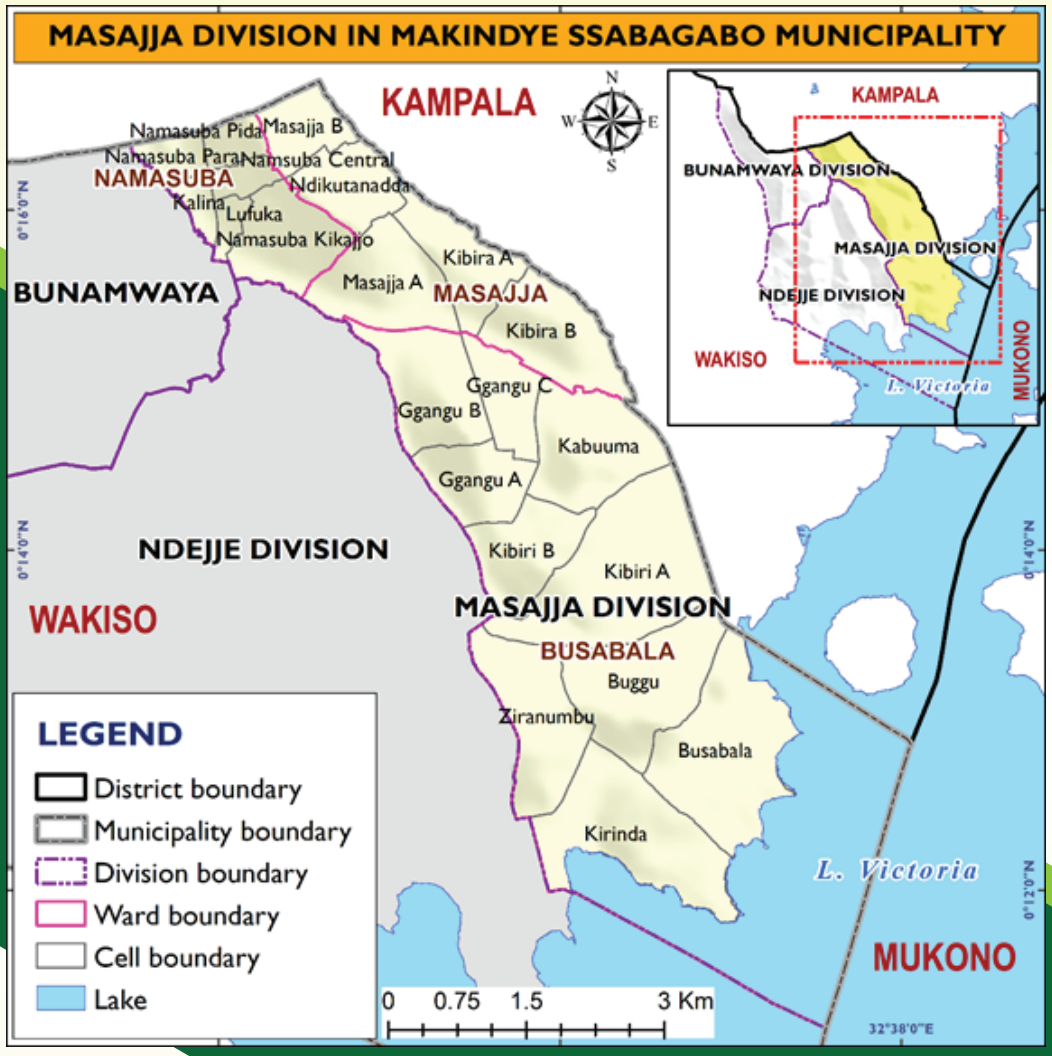
7. Appendices

Appendix 1: Detailed Costing of Actions

Result Area	Action	2026	2027	2028	2029	2030	Total Budget (UGX m)
Waste management improved	1. Enforce penalties for illegal dumping	48	60	48	48	36	240
	2. Establish waste material recovery facilities	120	150	120	120	90	600
	3. Develop municipal composting sites	144	180	144	144	108	720
	4. Introduce green fees/environmental levies	72	90	72	72	54	360
	5. Organize Monthly Recycling Awareness Drives	96	120	96	96	72	480
Water resource protection enhanced	1. Map and demarcate all wetland boundaries	66	82.5	66	66	49.5	330
	2. Enforce bylaws against wetland encroachment	88	110	88	88	66	440
	3. Promote sustainable wetland use	88	110	88	88	66	440
	4. Identify/map critical water sources	44	55	44	44	33	220
	5. Regulate car washing bays	66	82.5	66	66	49.5	330
	6. Promote rainwater harvesting	88	110	88	88	66	440
Urban greening & biodiversity	1. Conduct tree cover assessment	40	50	40	40	30	200
	2. Launch municipal tree-planting campaigns	100	125	100	100	75	500
	3. Develop community woodlots	100	125	100	100	75	500
	4. Map biodiversity hotspots	40	50	40	40	30	200

	5.	Raise awareness/enforce laws	60	75	60	60	45	300
	6.	Promote urban biodiversity	60	75	60	60	45	300
Pollution control	1.	Install air quality monitoring stations	90	112.5	90	90	67.5	450
	2.	Regulate emissions	75	93.75	75	75	56.25	375
	3.	Promote cleaner fuels & non-motorized transport	75	93.75	75	75	56.25	375
	4.	Control burning of waste	60	75	60	60	45	300
Climate resilience	1.	Develop Climate Change Adaptation Strategy	60	75	60	60	45	300
	2.	Promote energy-efficient technologies	60	75	60	60	45	300
	3.	Integrate climate-smart agriculture	60	75	60	60	45	300
	4.	Develop/update Disaster Risk Management Plan	40	50	40	40	30	200
	5.	Establish early warning systems	60	75	60	60	45	300
	6.	Conduct disaster preparedness drills	40	50	40	40	30	200
	7.	Build community disaster response committees	40	50	40	40	30	200
	8.	Develop post-disaster recovery plan	40	50	40	40	30	200
Community awareness & capacity building	1.	Conduct institutional audit	30.4	38	30.4	30.4	22.8	152
	2.	Create Environment Management Department	30.4	38	30.4	30.4	22.8	152
	3.	Integrate environmental focal points	30.4	38	30.4	30.4	22.8	152
	4.	Recruit environmental officers	38	47.5	38	38	28.5	190
	5.	Conduct in-service training	38	47.5	38	38	28.5	190
	6.	Develop career progression paths	22.8	28.5	22.8	22.8	17.1	114
	7.	Establish Environmental Governance Committee	22.8	28.5	22.8	22.8	17.1	114
	8.	Form Community Environmental Watch Committees	22.8	28.5	22.8	22.8	17.1	114
	9.	Develop partnerships with CSOs	30.4	38	30.4	30.4	22.8	152

10. Create dedicated Environment Fund	30.4	38	30.4	30.4	22.8	152
11. Mobilize external funding	30.4	38	30.4	30.4	22.8	152
12. Review/harmonize bylaws	22.8	28.5	22.8	22.8	17.1	114
13. Develop MSMC Local Environment Policy	22.8	28.5	22.8	22.8	17.1	114



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